Core practical – The combustion of alcohols

Pearson Edexcel GCSE (9-1) Chemistry

Name _____ **Class** _____ **Date** _____

Practical Objective

To investigate the temperature rise produced in a known mass of water by the combustion of the alcohols ethanol, propanol, butanol and pentanol.

Apparatus

- eye protection
- 100 cm^3 measuring cylinder
- 250 cm^3 conical flask
- stand, boss and clamp
- heat-resistant mat
- thermometer
- insulation/draught shield
- electronic balance
- spirit burners with caps containing: ethanol, propanol, butanol and pentanol

Safety

- Wear eye protection at all times.
- All alcohols are flammable: handle with care and keep the tops on burners when not in use.

Your teacher may watch to see if you can:

- keep careful control of control variables.

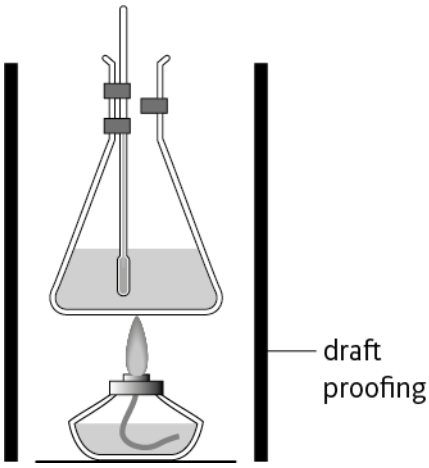
Method

A Measure the mass of an alcohol burner and cap. Record the mass and the name of the alcohol.

B Place the alcohol burner in the centre of a heat-resistant mat.

C Use a measuring cylinder to add 100 cm^3 of cold water to a conical flask.

D Measure and record the initial temperature of the water and carefully clamp the flask above the alcohol burner.



E Surround the apparatus with a draught screen/insulation.

F Remove the cap from the burner, then light the wick of the burner and allow the water to heat up by around 40 °C.

G Replace the cap on the burner and measure and record the final temperature of the water.

H Measure the mass of the alcohol burner and cap again and record the mass.

I Wash out the flask with cold water and repeat steps **A** to **H** using fresh cold water and a different alcohol.

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Recording your results

1 Record all your measurements in the table below.

Alcohol	Mass of burner + lid at start (g)	Mass of burner + lid at end (g)	Temperature of water at start (˚C)	Temperature of water at end (˚C)
ethanol				
propanol				
butanol				
pentanol				

Considering your results/conclusions

2 For each alcohol:

a Calculate the change in mass of the burner and lid. Fill in your answers in the table below.

b Calculate the change in temperature of the water. Fill in your answers in the table below.

c Divide the mass of each alcohol burnt by the change in temperature. This gives you the mass of each alcohol burnt to produce a 1°C rise in temperature. Fill in your answers in the table below.

Alcohol	Mass of alcohol burnt (g)	Temperature rise in water (˚C)	Mass of alcohol needed to produce a 1˚C rise in temperature
ethanol			
propanol			
butanol			
pentanol			

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3 **a** Describe any trends that you can see in your results.

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b Explain why these alcohols show such a regular trend in these values.
(*Hint:* think about molecular structures.)

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Evaluation

4 State four possible sources of error in this experiment and explain which is the main source of error.

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